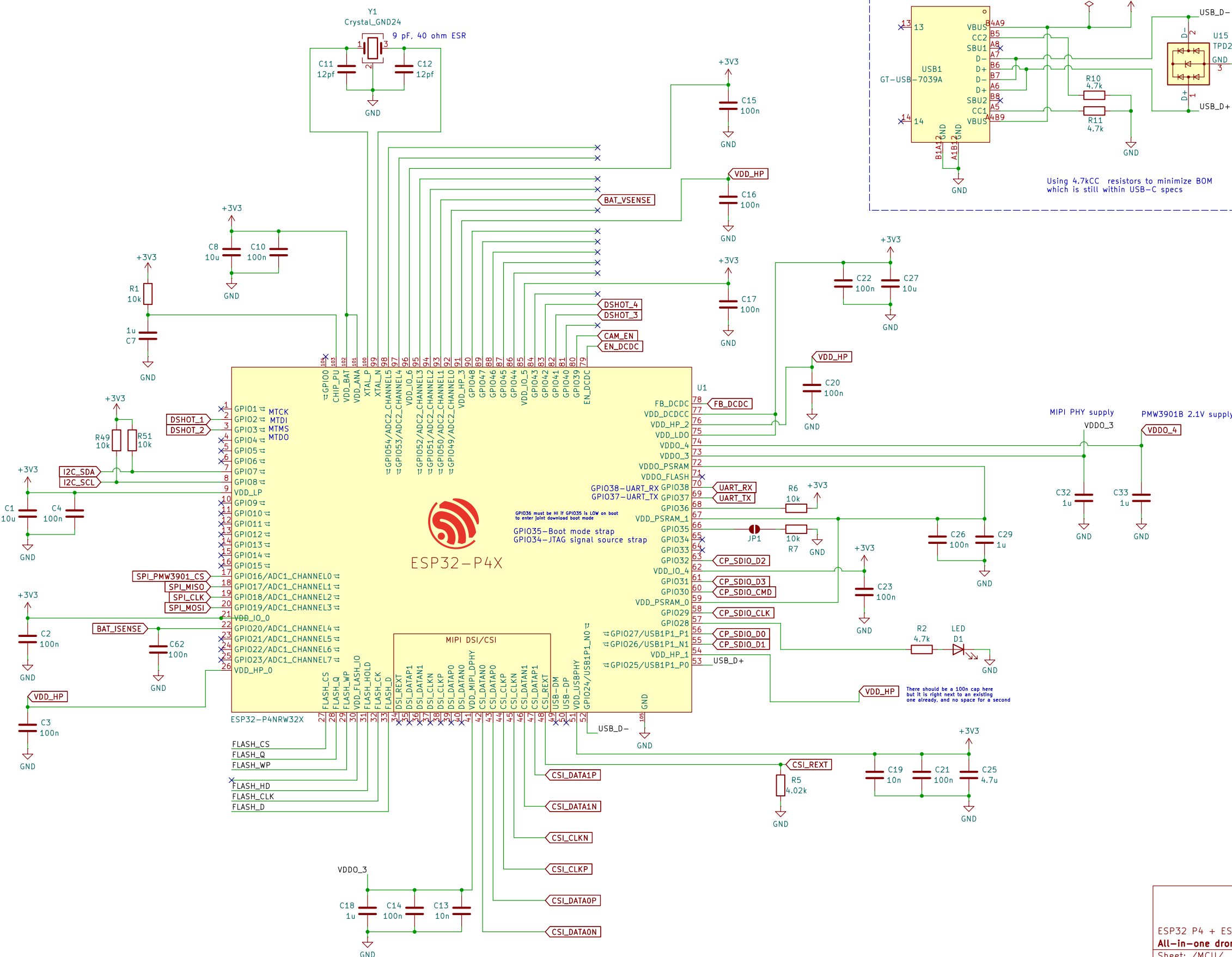
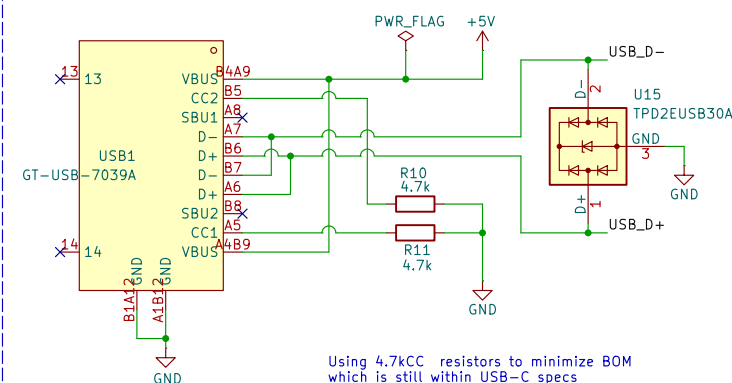


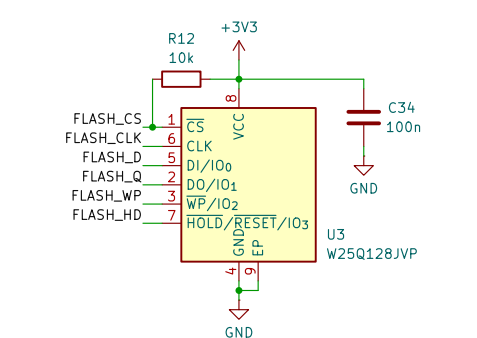
ESP32 P4 Flight Controller



USB

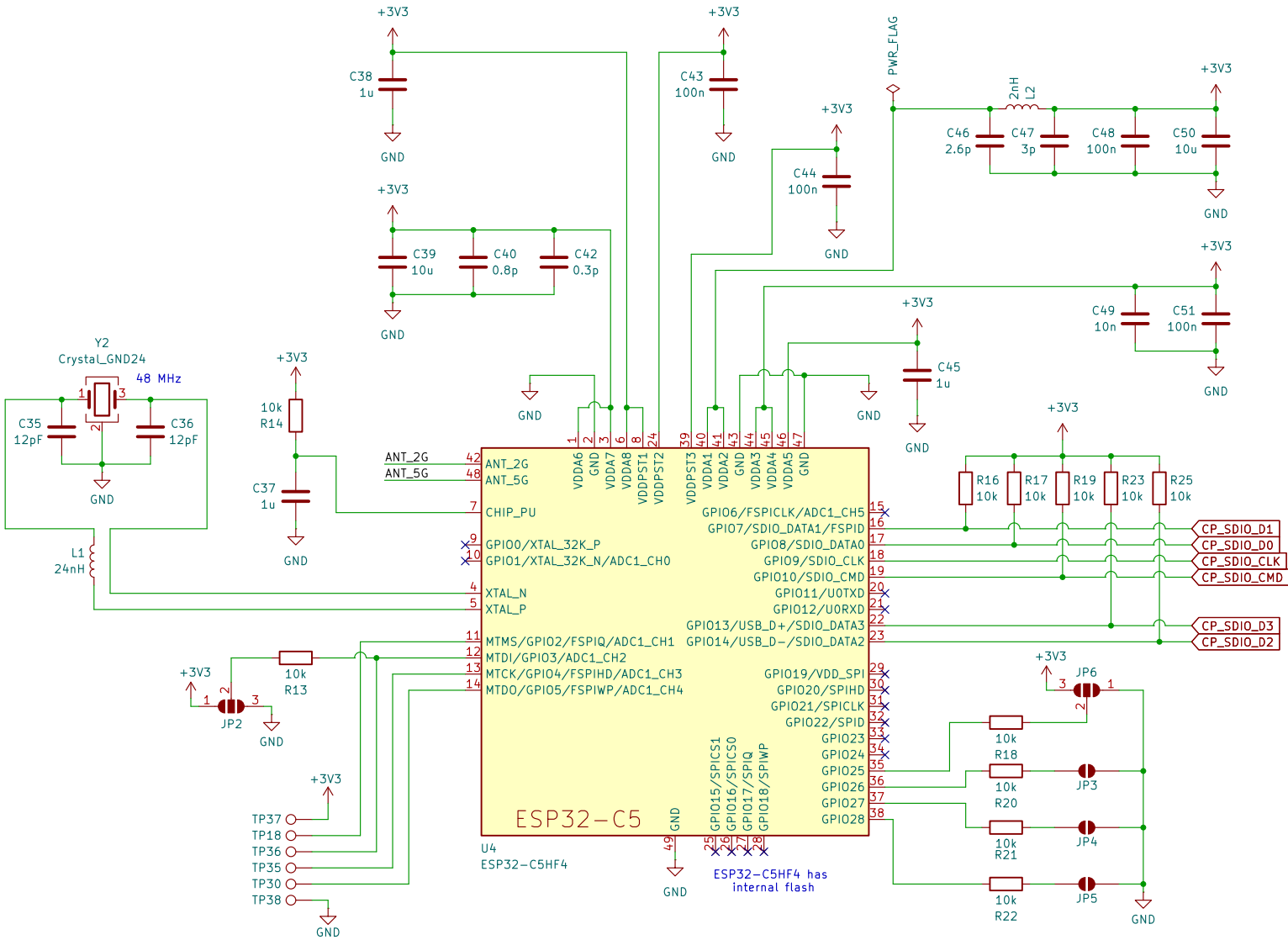


Flash



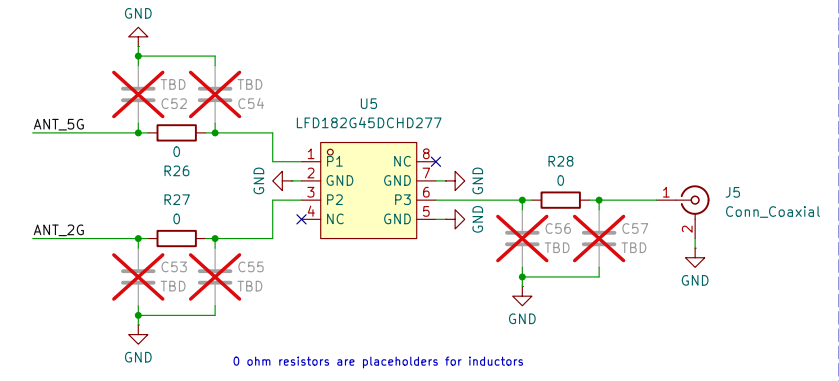
ESP32 P4 + ESP32 C5, 4xAM32 ESCs, 15-20 A per motor		
All-in-one drone flight controller, camera and 2S-3S ESC		
Sheet: /MCU/		
File: p4eregrine.kicad_sch		
Title: P4eregrine		
Size: A3	Date: 2026-09-22	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/9

ESP32 C5 Wireless Coprocessor



Boot mode	SPI boot	JTAG download	SDIO download
GPIO 26	X	X	0
GPIO 27	X	1	0
GPIO 28	1	0	0
27 & 28 have internal pullups			

RF 2.4 GHz + 5 GHz



ESP32 P4 + ESP32 C5, 4xAM32 ESCs, 15–20 A per motor
All-in-one drone flight controller, camera and 2S–3S ESC
Sheet: /Coprocessor/
File: coprocessor.kicad_sch

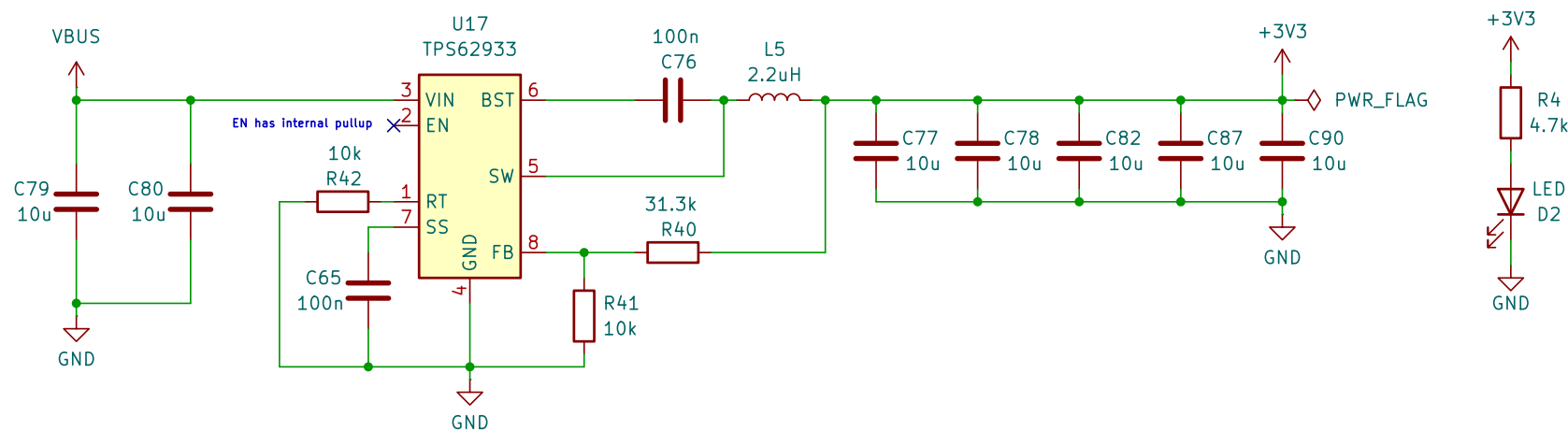
Title: P4eregrine

Size: A3 Date: 2026–09–22
KiCad E.D.A. 10.0.6

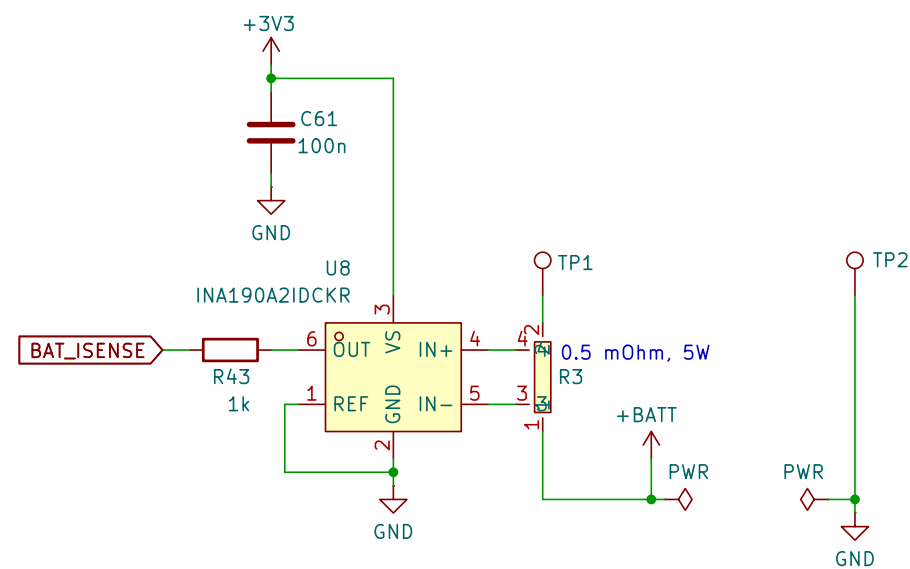
Rev: A
Id: 2/9

Buck 3V3 Converter

Maximum expected continuous current draw: 1.5 A
Maximum 3V3 supply capability at 3S: 2.3 A (inductor limit)

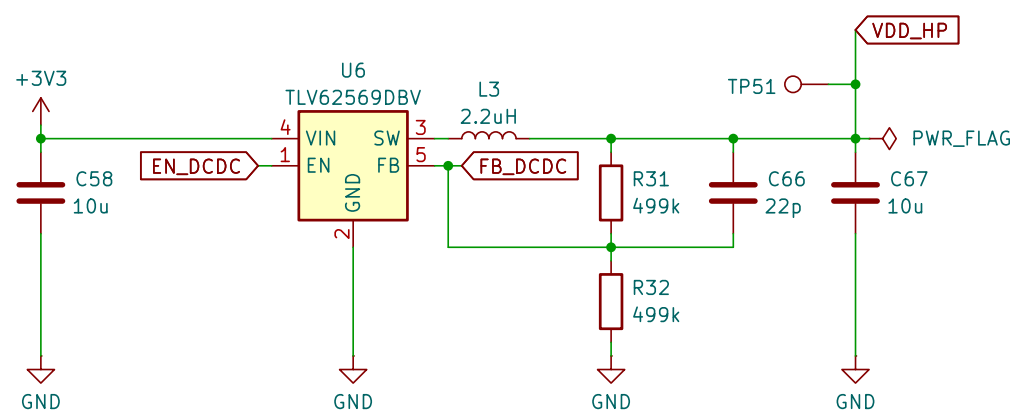


Battery Measurement



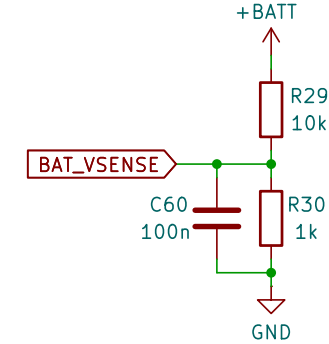
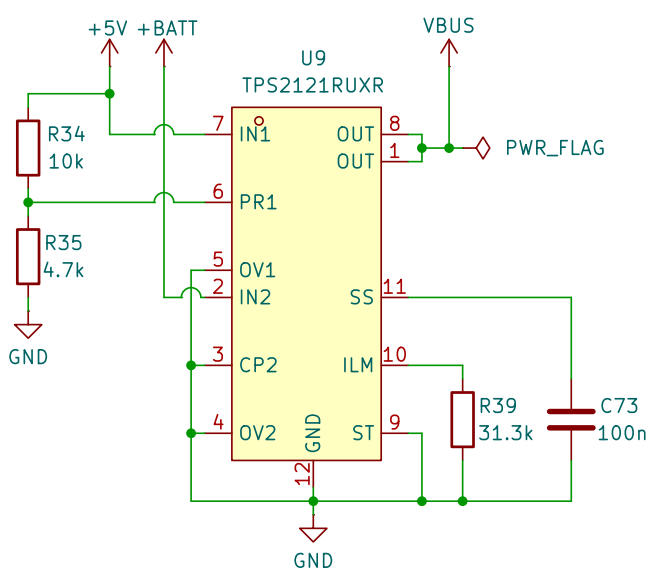
ESP32 P4 HP DC-DC converter

Only used by ESP32 P4



USB/Battery Power MUX

The 3V3 converter is fed from USB 5V with priority;
when it drops, +BAT is routed to VBUS instead



ESP32 P4 + ESP32 C5, 4xAM32 ESCs, 15-20 A per motor
All-in-one drone flight controller, camera and 2S-3S ESC

Sheet: /Power/
File: power.kicad_sch

Title: P4eregrine

Size: A4

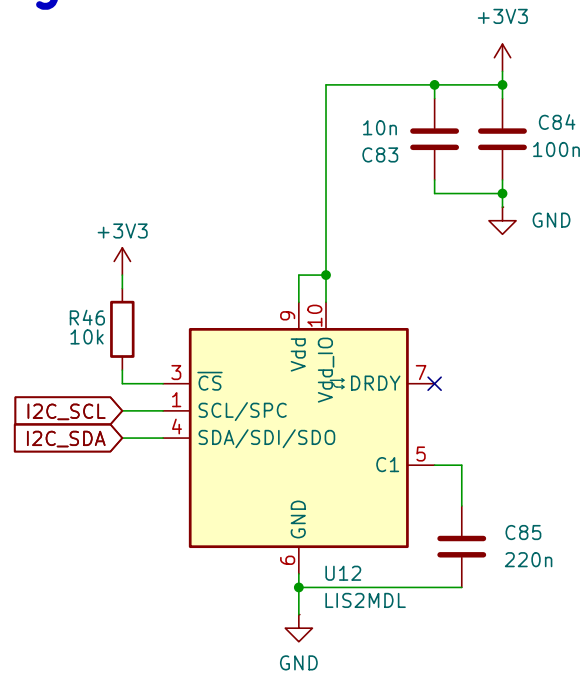
Date: 2026-09-22

Rev: A

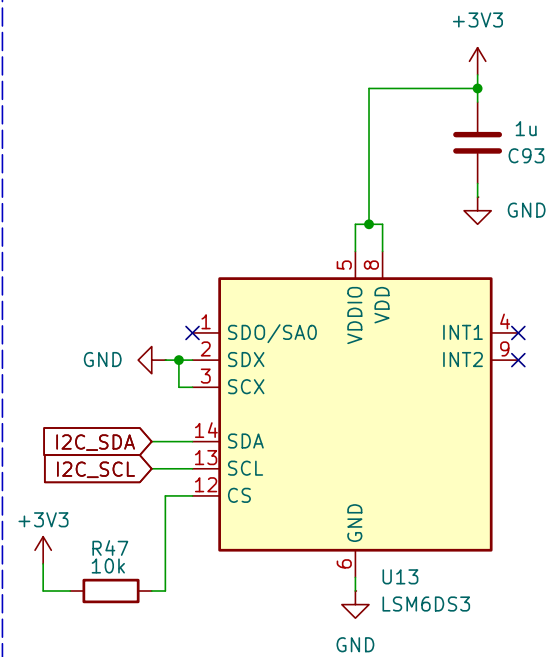
KiCad E.D.A. 10.0.6

Id: 3/9

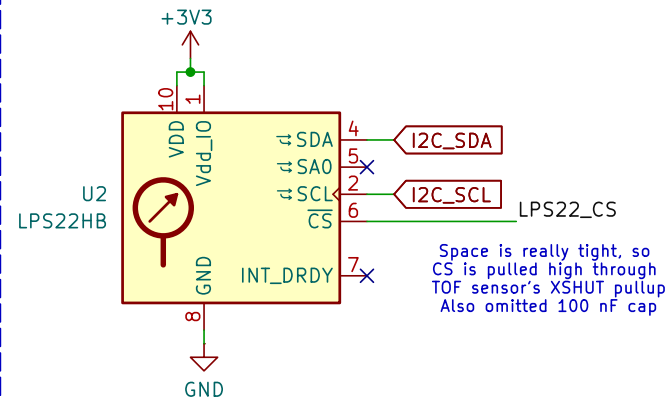
Magnetometer



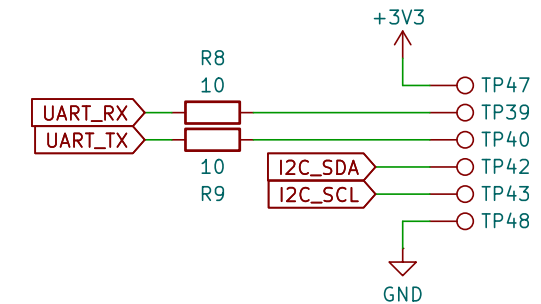
IMU



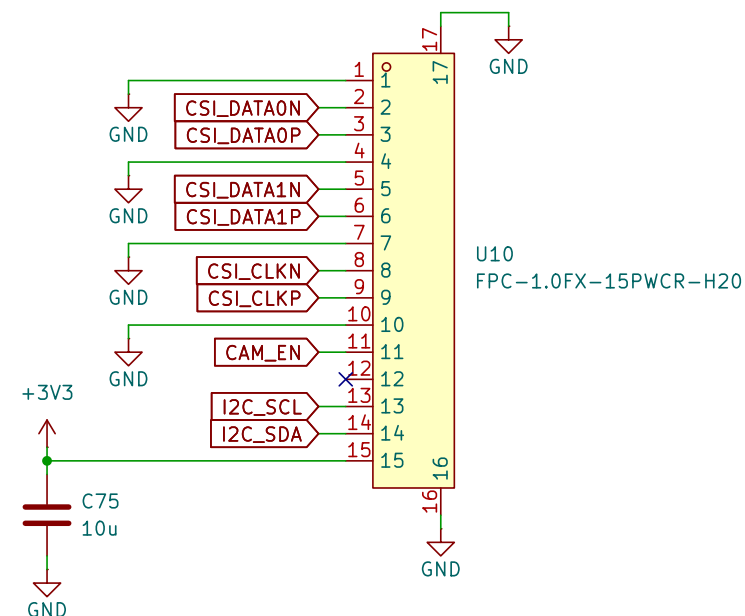
Barometer



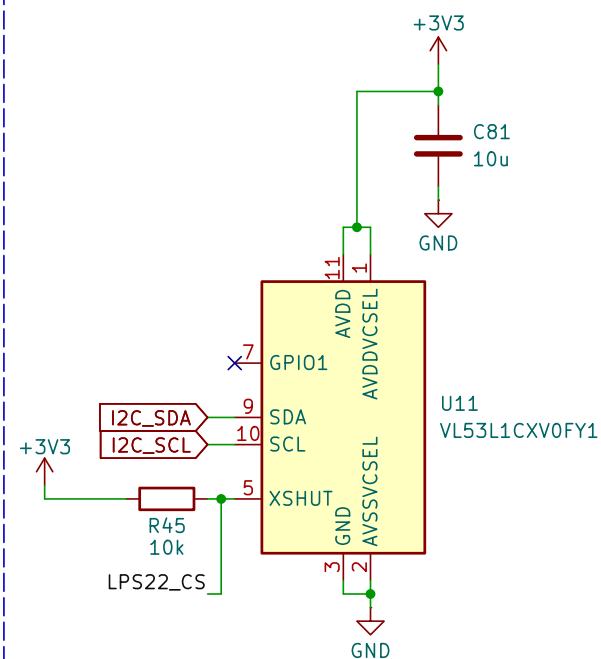
UART, I2C, GPIO breakout



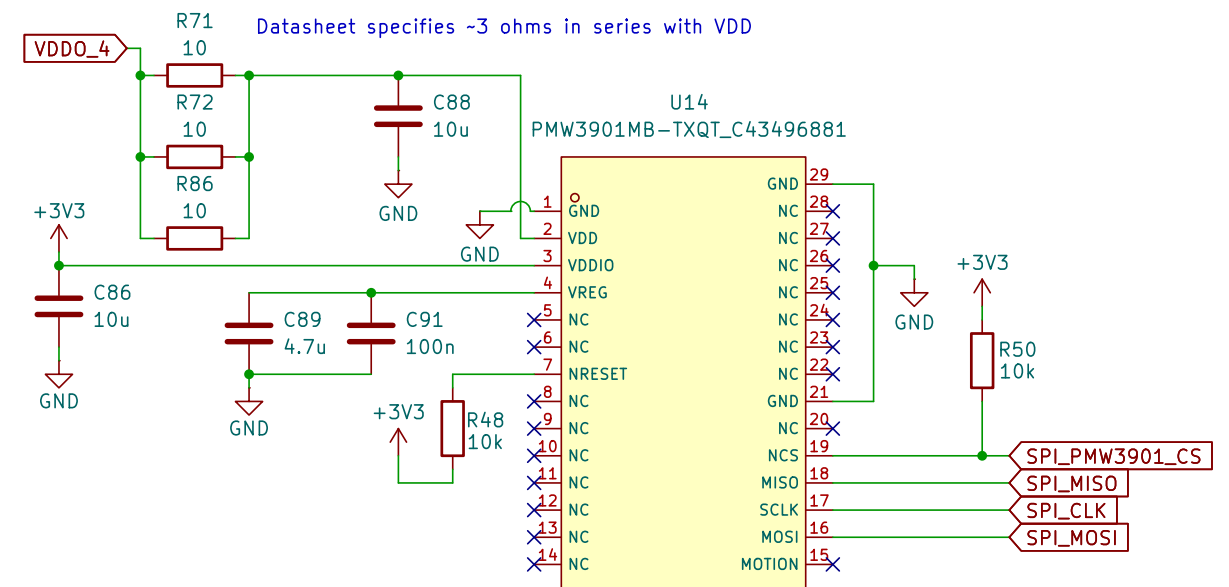
RPi 15-pin Camera Connector



ToF Sensor



Optical Flow Sensor



ESP32 P4 + ESP32 C5, 4xAM32 ESCs, 15–20 A per motor
All-in-one drone flight controller, camera and 2S–3S ESC

Sheet: /Peripherals/
File: peripherals.kicad_sch

Title: P4eregrine

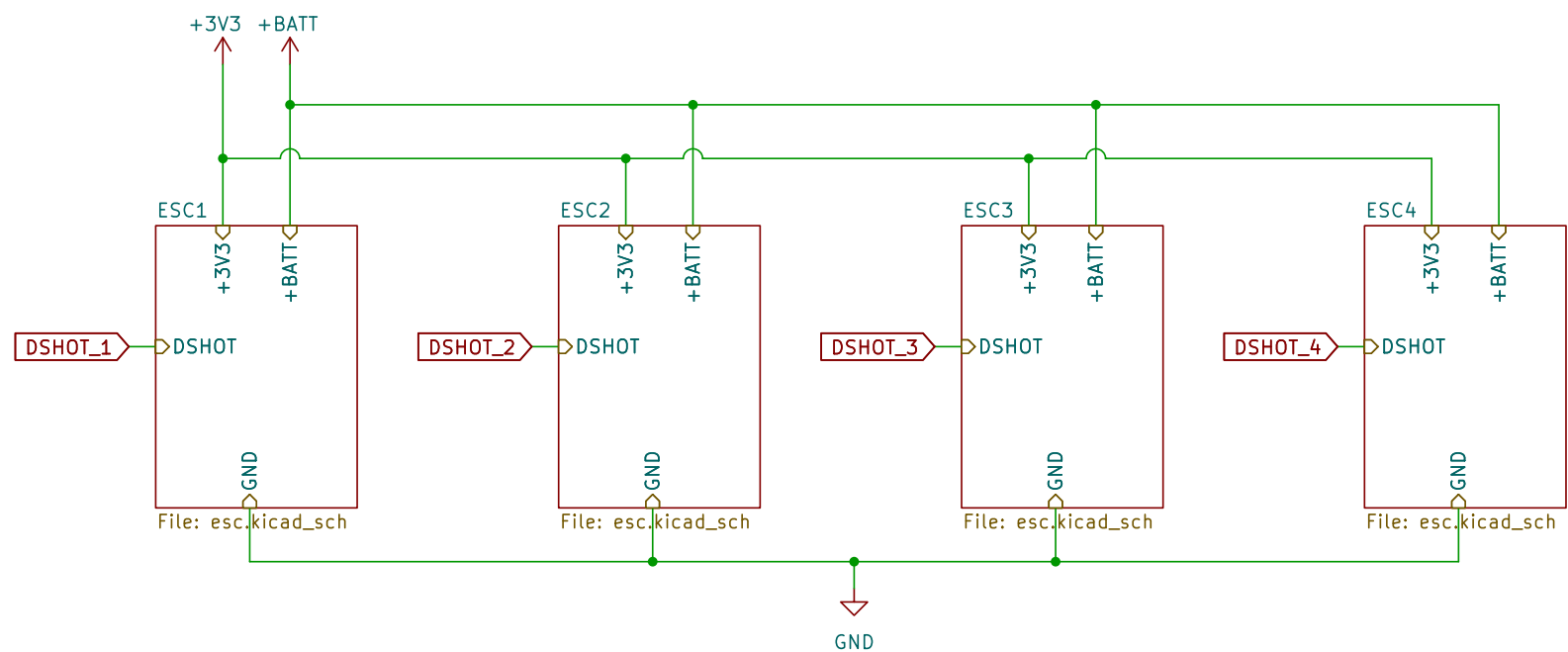
Size: A4	Date: 2026-09-22
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SIZE: A1	
KiCad E.D.A. 10.0.6	

Rev: A

Id: 4/9

Brushless Motor Electronic Speed Control



ESP32 P4 + ESP32 C5, 4xAM32 ESCs, 15–20 A per motor
All-in-one drone flight controller, camera and 2S–3S ESC

Sheet: /Motors/
File: motors.kicad_sch

Title: P4eregrine

Size: A4

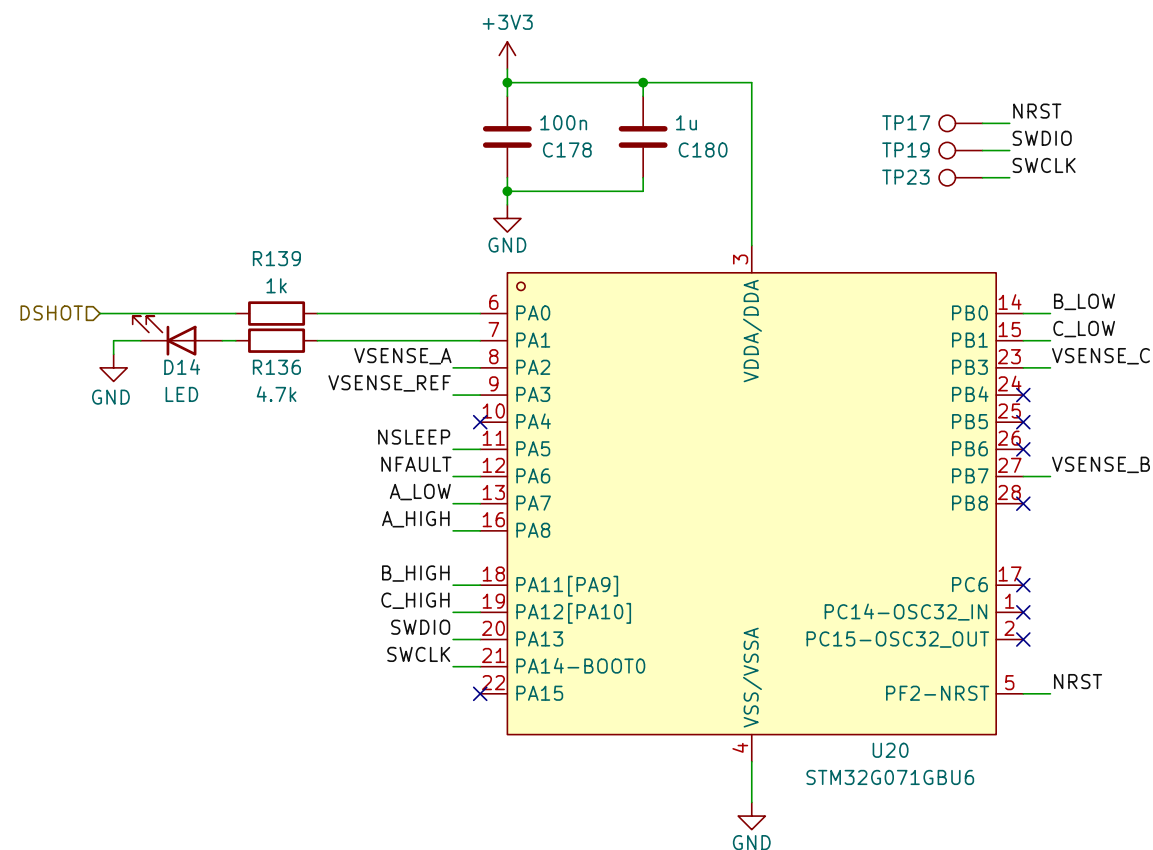
Date: 2026-09-22

Rev: A

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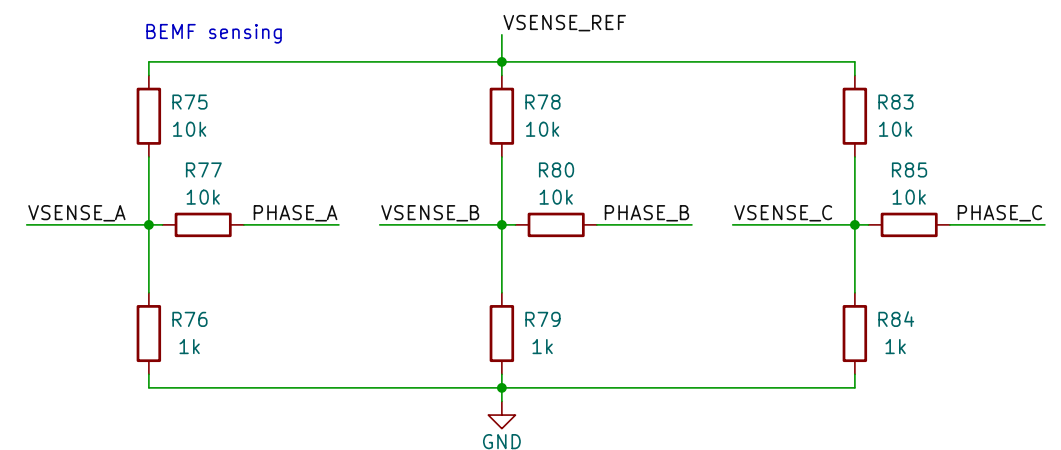
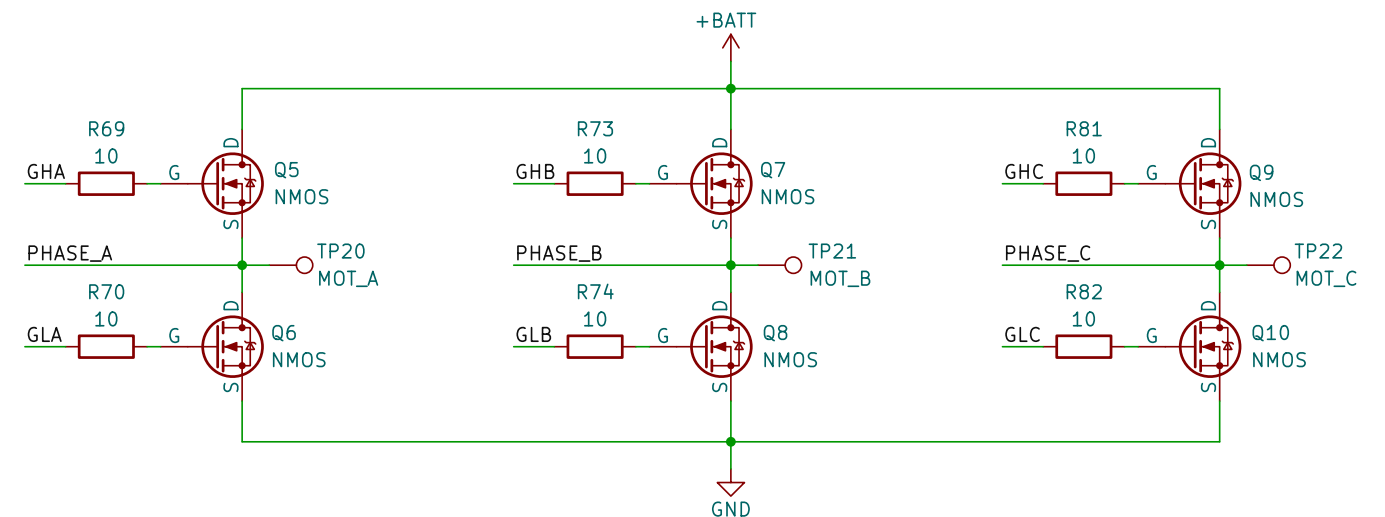
Id: 5/9

MCU



MOSFETs

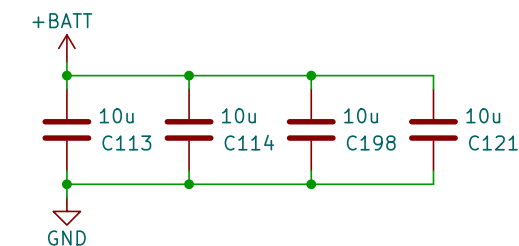
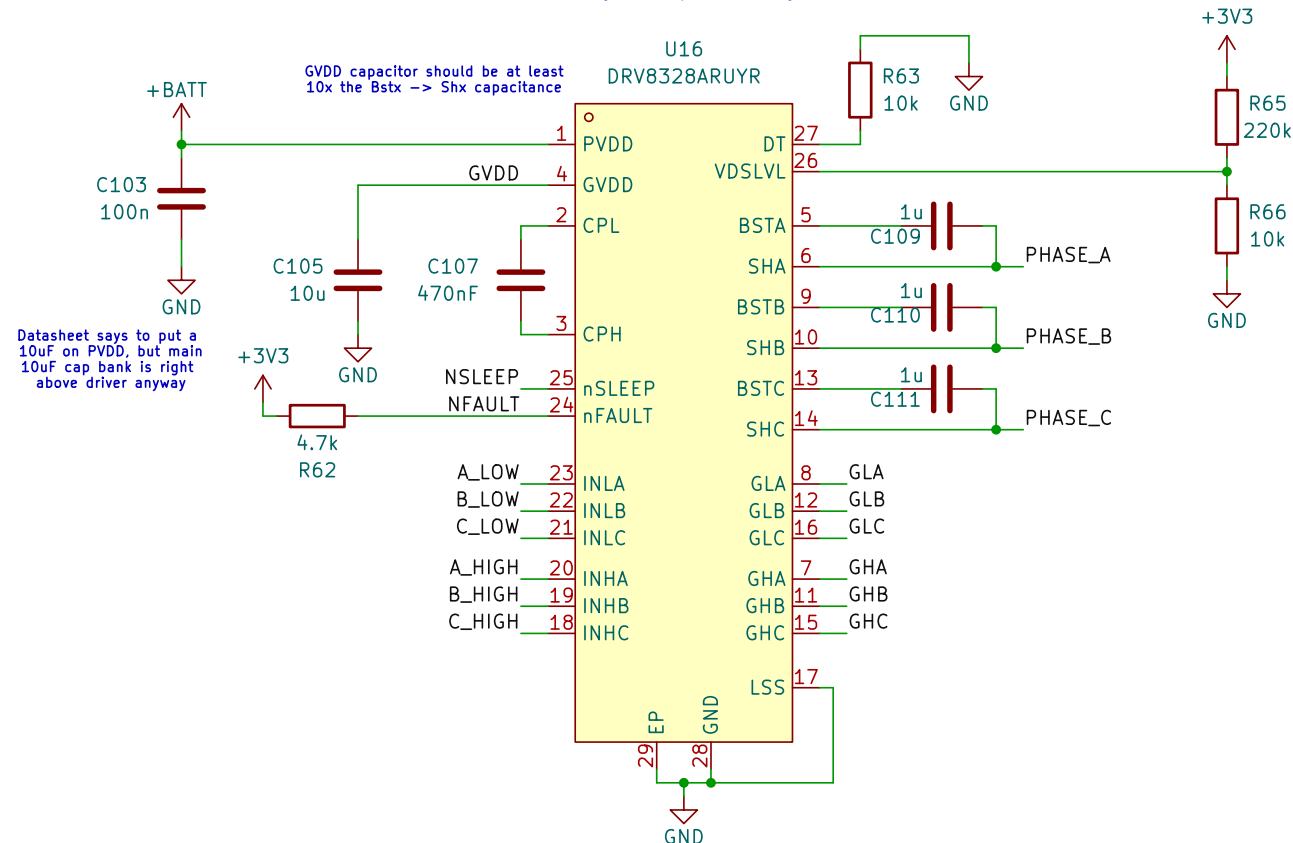
JMSL0302AU, 2.4 mOhm@10V at 100 deg C



Gate driver

$R_{dt} = \text{deadtime(ns)}/5 - 10k; -100 \text{ ns with } 10k$
This is a minimum value, if a different deadtime is generated by the MCU, the greater of the two will be used

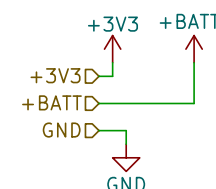
$3.3 * 10 / (220+10) = 0.1435 \text{ V drop}$
Assuming 2.4 mOhms at 100 deg C, trip current is ~60 A



1210 10uF CL32B106KAJNNNE caps retain ~7 uF even at 3S HV, which results in ~28 uF per ESC in the worst case scenario.

Alternatively, CL32Y226KAVVPN 22 uF caps retain ~15 uF when derated, but they cost way more.

Dummy labels so power inputs show up on the hierarchical schematic outline



ESP32 P4 + ESP32 C5, 4xAM32 ESCs, 15-20 A per motor
All-in-one drone flight controller, camera and 2S-3S ESC

Sheet: /Motors/ESC1/
File: esc.kicad_sch

Title: P4eregrine

Size: A4

Date: 2026-09-22

Rev: A

KiCad E.D.A. 10.0.6

Id: 6/9